

SQL Filtering Data

04 · Filtering Data — Complete Reference Guide

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1. Comparison Operators — = <> > >= < <=

→ *Equal to: Retrieve all customers from Germany*

```
SELECT *  
FROM customers  
WHERE country = 'Germany'
```

→ *Not equal to: Retrieve all customers NOT from Germany*

```
SELECT *  
FROM customers  
WHERE country <> 'Germany'
```

→ *Greater than: score > 500*

```
SELECT *  
FROM customers  
WHERE score > 500
```

→ *Greater than or equal to: score >= 500*

```
SELECT *  
FROM customers  
WHERE score >= 500
```

→ *Less than: score < 500*

```
SELECT *  
FROM customers
```

```
WHERE score < 500
```

→ *Less than or equal to: score <= 500*

```
SELECT *  
FROM customers  
WHERE score <= 500
```

2. Logical Operators — AND, OR, NOT

→ *AND: customers from USA with score > 500*

```
SELECT *  
FROM customers  
WHERE country = 'USA' AND score > 500
```

→ *OR: customers from USA OR with score > 500*

```
SELECT *  
FROM customers  
WHERE country = 'USA' OR score > 500
```

→ *NOT: customers whose score is NOT less than 500*

```
SELECT *  
FROM customers  
WHERE NOT score < 500
```

3. Range Filtering — BETWEEN

→ *Retrieve customers with score between 100 and 500 (inclusive)*

```
SELECT *  
FROM customers  
WHERE score BETWEEN 100 AND 500
```

→ *Alternative method — equivalent to BETWEEN*

```
SELECT *  
FROM customers  
WHERE score >= 100 AND score <= 500
```

4. Set Filtering — IN

→ *Retrieve customers from Germany OR USA*

```
SELECT *  
FROM customers  
WHERE country IN ('Germany', 'USA')
```

5. Pattern Matching — LIKE

→ *First name starts with 'M' → use % after the letter*

```
SELECT *  
FROM customers  
WHERE first_name LIKE 'M%'
```

→ *First name ends with 'n' → use % before the letter*

```
SELECT *  
FROM customers  
WHERE first_name LIKE '%n'
```

→ *First name contains 'r' → use % on both sides*

```
SELECT *  
FROM customers  
WHERE first_name LIKE '%r%'
```

→ *'r' is in the third position → use _ for each unknown character*

```
SELECT *  
FROM customers  
WHERE first_name LIKE '__r%'
```